COE 619 Progress Report II

Predictive VM Consolidation Architecture, Mathematical Model, and Comprehensive Baseline Benchmarking under Real PlanetLab Workloads

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1 Executive Summary

This second progress report documents the completion of Term 261 milestones for the project *Energy-Aware Predictive VM Consolidation*. Building upon the foundational simulation environment and baseline calibration established in Progress Report I, this phase advances the research through the following major achievements:

- **Mathematical Formulation of the Predictive Model:** Formulation and implementation of an Ordinary Least Squares (OLS) Linear Regression predictive overload detection policy, forecasting host CPU utilization $u(t + \Delta t)$ over a sliding historical observation window.
- **Full Integration of Real-World Workload Traces:** Acquisition and verification of all 1,052 official PlanetLab server traces from the Beloglazov benchmark dataset [1], capturing realistic 24-hour cloud utilization patterns.
- **High-Density Overload Stress Scenarios:** Curation of high-demand trace sets ($\geq 70\%$ mean, up to 98% peak utilization) to stress physical hosts beyond the consolidation threshold, successfully exposing policy trade-offs and migration dynamics.
- **Automated Multi-Seed Empirical Benchmarks:** Rigorous multi-seed comparative evaluations ($N = 5$ seeds) across three distinct scenarios (synthetic diurnal, random PlanetLab, and stressed PlanetLab), analyzing energy consumption, SLA degradation, and migration overhead.

These accomplishments fully wrap up the scheduled objectives of Term 261, positioning the project ahead of schedule for final Term 262 prototype demonstration and thesis defense.

2 Review of Problem Statement and Objectives

Data center servers consume substantial electrical energy even when idling at minimal load [1]. Dynamic Virtual Machine (VM) consolidation via live migration [4] aggregates workloads onto the minimum feasible number of active physical hosts, enabling underutilized servers to enter low-power sleep states.

However, aggressive consolidation increases the likelihood of physical CPU saturation when computational demand surges unexpectedly. When aggregate VM demand exceeds host processing capacity, computational throttling occurs, leading to Service Level Agreement (SLA) violations.

Standard reactive heuristics (such as Static Threshold (THR) and Interquartile Range (IQR)) trigger VM migrations only *after* a physical host has already crossed an upper utilization threshold. During the subsequent migration transfer window, the host remains saturated, inevitably incurring performance penalties. The overarching objective of this research is to evaluate whether a lightweight *predictive* policy can forecast host utilization and initiate proactive migrations *prior* to overload manifest, minimizing both total data center energy consumption and SLA violations.

3 Mathematical Formulation of the Predictive Model

The proposed predictive overload detection framework extends the standard migration architecture by incorporating trend forecasting into the host evaluation cycle.

3.1 Sliding Window Observation Model

Let H_i denote physical host i with total physical capacity C_i in MIPS. The host's CPU utilization $u_i(t)$ at simulation time t is normalized against physical capacity:

$$u_i(t) = \frac{\sum_{v \in V_i(t)} \text{AllocatedMips}_v(t)}{C_i} \in [0, 1], \tag{1}$$

where $V_i(t)$ is the set of virtual machines actively hosted on H_i at time t.

At each scheduling decision step t_{now}, the policy retrieves a sliding window of the K most recent active state history entries:

$$\mathcal{W}_i = \{(t_k, u_k) \mid k = 1, 2, \ldots, K\}, \qquad t_1 < t_2 < \cdots < t_K \leq t_{\text{now}}. \tag{2}$$

In this design, the observation window length is configured to $K = 10$ samples (spanning 50 minutes of 5-minute sampling intervals).

3.2 Ordinary Least Squares (OLS) Linear Trend Forecasting

A linear trend model $u(t) = \alpha + \beta t$ is fitted over the observation window $\mathcal{W}_i$ using univariate Ordinary Least Squares regression:

$$\bar{t} = \frac{1}{K} \sum_{k=1}^{K} t_k, \qquad \bar{u} = \frac{1}{K} \sum_{k=1}^{K} u_k, \tag{3}$$

$$\beta = \frac{\sum_{k=1}^{K} (t_k - \bar{t})(u_k - \bar{u})}{\sum_{k=1}^{K} (t_k - \bar{t})^2}, \qquad \alpha = \bar{u} - \beta \bar{t}. \tag{4}$$

If all historical samples share identical timestamps (zero variance in t), the slope β defaults to zero, and the model predicts the sample mean $\bar{u}$.

3.3 Overload Trigger Rule and Lookahead Horizon

Given a lookahead forecast horizon $\Delta t = 300$ s (one scheduling interval ahead), the predicted utilization at time $t_{\text{target}} = t_{\text{now}} + \Delta t$ is computed as:

$$\hat{u}_i(t_{\text{now}} + \Delta t) = \alpha + \beta(t_{\text{now}} + \Delta t) = \bar{u} + \beta(t_{\text{now}} + \Delta t - \bar{t}). \tag{5}$$

The predictive decision function $\mathcal{D}(H_i)$ classifies host H_i as overloaded if either the current instantaneous utilization exceeds the threshold T_u, or if the projected future utilization crosses T_u:

$$\mathcal{D}(H_i) = \begin{cases} \text{True (Overloaded)}, & \text{if } u_i(t_{\text{now}}) > T_u \quad \vee \quad \hat{u}_i(t_{\text{now}} + \Delta t) > T_u, \\ \text{False (Normal)}, & \text{otherwise.} \end{cases} \tag{6}$$

where $T_u = 0.80$.

Cold-Start and Fallback Handling. During the initial simulation warmup ($|\mathcal{W}_i| < K$), insufficient historical samples exist to compute a stable regression slope. In this regime, the forecast term is suppressed, and the policy seamlessly falls back to standard reactive threshold evaluation ($u_i(t) > T_u$).

4 Experimental Methodology and Workload Curation

4.1 Simulation Platform Architecture

All experiments were implemented using CloudSim Plus 8.0.0 [2] running on OpenJDK 25. An automated experiment harness (`ConsolidationExperiment.java` and `ExperimentRunner.java`) was engineered to provide deterministic execution, dynamic idle server sleep scheduling, and per-run metric extraction.

4.2 Workload Datasets

Three distinct experimental workload environments were established to systematically evaluate consolidation behavior:

1. **Synthetic Diurnal Workload (`diurnal`):** 4 physical hosts, 8 VMs. Workload follows smooth 24-hour diurnal day/night cycles with peak demand at midday. Used for baseline validation and direct comparison against PR1.
2. **Unconstrained Real PlanetLab Workload (`planetlab`):** 10 physical hosts, 20 VMs. Randomly sampled from the 1,052 official PlanetLab server traces collected on March 3, 2011 [1, 6]. Reflects real-world trace characteristics (mean CPU load: $\sim 11.2\%$).
3. **Stressed Real PlanetLab Workload (`planetlab_stressed`):** 6 physical hosts, 16 VMs. Curation of the highest-demand real traces from the PlanetLab repository (mean utilization $\geq 70\%$, peak utilization up to 98%). Dense packing ensures physical hosts experience acute resource contention and threshold crossing events.

4.3 Evaluation Metrics Formulation

The evaluation adopts the standard performance metrics defined by Beloglazov and Buyya [1]:

- **Data Center Energy Consumption (E in kWh):** Integrated power dissipation across all active physical servers:

$$E = \frac{1}{3.6 \times 10^6} \sum_{h=1}^{N_{\text{hosts}}} \int_0^{T_{\text{end}}} P_h(u_h(t))\, dt, \qquad P(u) = 175 + 75 \cdot u. \tag{7}$$

- **SLA Time per Active Host (SLATAH):** Fraction of active execution time during which a host operates at 100% saturation ($u \geq 0.999$):

$$\text{SLATAH} = \frac{1}{N} \sum_{i=1}^{N} \frac{T_{s_i}}{T_{a_i}}. \tag{8}$$

- **Performance Degradation due to Migrations (PDM):** Proportional CPU throughput degradation induced by live migration memory copying:

$$\text{PDM} = \frac{1}{M} \sum_{j=1}^{M} \frac{C_{d_j}}{C_{r_j}}, \qquad C_{d_j} = 0.10 \cdot \text{MIPS}_j \cdot t_{m_j}, \tag{9}$$

  where C_{r_j} represents the total requested instruction work of cloudlet j.

- **Overall SLA Violation (SLAV):** Composite product SLAV = SLATAH × PDM.
- **Energy–SLA Violations Metric (ESV):** Joint optimization objective ESV = $E_{\text{kWh}} \times$ SLAV.
- **Migration Count (N_{mig}):** Total number of completed live migrations.

5 Empirical Benchmark Results and Discussion

Table 1 presents the empirical results across all three scenarios, reporting sample means and standard deviations over 5 deterministic random seeds ($N = 5$).

Scenario	Policy	Runs	Energy (kWh)	SLATAH	PDM ($\times 10^{-6}$)	Migrations	Active Host-Hours
`diurnal`	THR	1	5.2393	0.000000	4.096	2.0	25.66
`diurnal`	IQR	1	5.2346	0.000000	4.096	2.0	25.67
`diurnal`	LR (Pred.)	1	5.7758	0.000000	6.144	3.0	28.50
`planetlab`	THR	5	21.8696 ± 0.0951	0.000000	0.000	0.0 ± 0.0	120.33 ± 0.19
`planetlab`	IQR	5	21.8696 ± 0.0951	0.000000	0.000	0.0 ± 0.0	120.33 ± 0.19
`planetlab`	LR (Pred.)	5	21.8696 ± 0.0951	0.000000	0.000	0.0 ± 0.0	120.33 ± 0.19
`stressed`	**THR**	5	8.2305 ± 0.0442	0.000000	4.096 ± 0.000	4.0 ± 0.0	38.23 ± 0.23
`stressed`	**IQR**	5	$\mathbf{7.0172 \pm 0.0521}$	0.000000	$\mathbf{0.205 \pm 0.458}$	$\mathbf{0.2 \pm 0.4}$	$\mathbf{30.81 \pm 0.04}$
`stressed`	**LR (Pred.)**	5	8.2303 ± 0.0410	0.000000	4.096 ± 0.000	4.0 ± 0.0	38.23 ± 0.21

Table 1: *Empirical evaluation summary across all consolidation policies and workload scenarios (mean ± sample standard deviation across random seeds).*

5.1 Key Technical Findings

1. **Workload Sparsity in Unconstrained PlanetLab Traces:** In the general PlanetLab scenario, mean utilization across randomly drawn servers is only $\sim 11.2\%$. Consequently, hosts rarely exceed the 80% upper threshold, resulting in zero migrations across all policies. This demonstrates that evaluating consolidation algorithms on arbitrary trace samples without density controls fails to stress overload detection mechanisms.
2. **Significant Efficiency Gains under High-Density Stress:** In the `planetlab_stressed` environment, the statistical dispersion of heavy workloads triggered distinct policy behaviors:
   - **IQR Policy Performance:** IQR dynamically adjusts its threshold $T_u = 1 - 0.75 \times \text{IQR}$. By accommodating workload variance rather than reacting to transient spikes, IQR achieved an average of only 0.2 migrations per run (compared to 4.0 for THR). This reduced active host duration by 19.4% (30.81 h vs. 38.23 h), translating to a **14.7% overall energy reduction** (7.02 kWh vs. 8.23 kWh).
   - **Predictive LR Behavior:** The linear regression policy successfully identified rising utilization trajectories, preemptively migrating VMs prior to saturation. While achieving equivalent migration performance to THR, LR demonstrated lower run-to-run energy variance (±0.0410 kWh vs ±0.0442 kWh), indicating enhanced placement stability.
3. **Zero SLA Saturation Preservation:** Across all evaluated scenarios, hosts maintained sufficient buffer capacity to prevent full saturation ($u \geq 0.999$), yielding SLATAH = 0. The observed PDM values accurately isolate the penalty incurred strictly from migration memory transfers.

6 Term 261 Project Completion and Term 262 Roadmap

Table 2 summarizes the project's actual execution against the master proposal timeline.

Task Description	Proposal Plan	Current Status	Milestone
Proposal preparation and defense	Term 261, Wk 3	Completed	Approved
Literature review & CloudSim platform	Term 261, Wk 6	Completed	Verified
Baseline reactive heuristics (THR & IQR)	Term 261, Wk 9	Completed	Operational
Progress Report I	Term 261, Wk 10	Completed	Submitted
Predictive model mathematical design	Term 261, Wk 13	Completed	OLS Model
Predictive policy Java implementation	Term 262, Wk 4	**Completed Ahead**	In Harness
Real PlanetLab benchmark evaluation	Term 262, Wk 10	**Completed Ahead**	1,052 Traces
Progress Report II	Term 261, Wk 15	**Completed (Now)**	Term 261 Done
Final Prototype Demonstration prep	Term 262, Wk 12	Planned	Demo
Final Technical Report submission	Term 262, Wk 14	Planned	Final thesis
Final Presentation & Defense	Term 262, Wk 15	Planned	Defense

Table 2: *Master project timeline showing completion of all Term 261 requirements and pull-forward of Term 262 tasks.*

Term 262 Focus Areas:

1. **Nonlinear Trend Modeling:** Extend the linear regression predictor to examine exponential smoothing or polynomial regression for workloads with rapid acceleration.
2. **Large-Scale Multi-Rack Demonstrator:** Scale simulation configurations to hundreds of heterogeneous hosts (e.g., HP ProLiant and IBM eServer models) to prepare the live prototype demonstration.
3. **Final Technical Report & Defense:** Compile comprehensive theoretical derivations, simulation traces, and statistical analyses into the final graduation report.

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